Review Test Submission: Recap Assignment

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Course	2021-ST311 Applied Statistics I
Test	Recap Assignment
Started	07/12/20 14:21
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Due Date	21/12/20 12:00
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Attempt Score	104 out of 110 points
Time Elapsed	48 hours, 45 minutes
Instructions	Answer ALL Questions. Statistical Tables are added here for your convenience. Standard_Normal_Tables.pdf F_distribution_tables.pdf t_distribution_tables.pdf • Feedback is not provided immediately on submission. Submissions are marked manually not automated. Feedback will be provided when all submissions have been graded. • While working on your submission you can click the SAVE button and return to your workings at a later time to continue (as long as it is before the due date). • To complete your submission, i.e. submit your workings for grading, click the SUBMIT button. • You should need to make only 1 submission. However, a maximum of 2 attempts is allowed, the 2nd attempt is a back up in case you inadvertently click submit before you were finished.
Results Displayed	Submitted Answers, Feedback, Incorrectly Answered Questions

Question 1

0 out of 1 points

In a study into typical monthly mobile revenue per user (RPU), random samples of mobile customers were selected, 10 prepay customers and 7 billpay customers.

The usage in euro observed for those individual customers last month shows a right-skewed distribution.

Whichof following tests would you apply:

Selected Answer:

Signs test for a median

Question 2

1 out of 1 points

The true correlation coefficient between two variables X and Y for a population of individuals is known to have value ρ = 0. This implies that X and Y are unrelated." Is this statement true or false?

Selected Answer: False

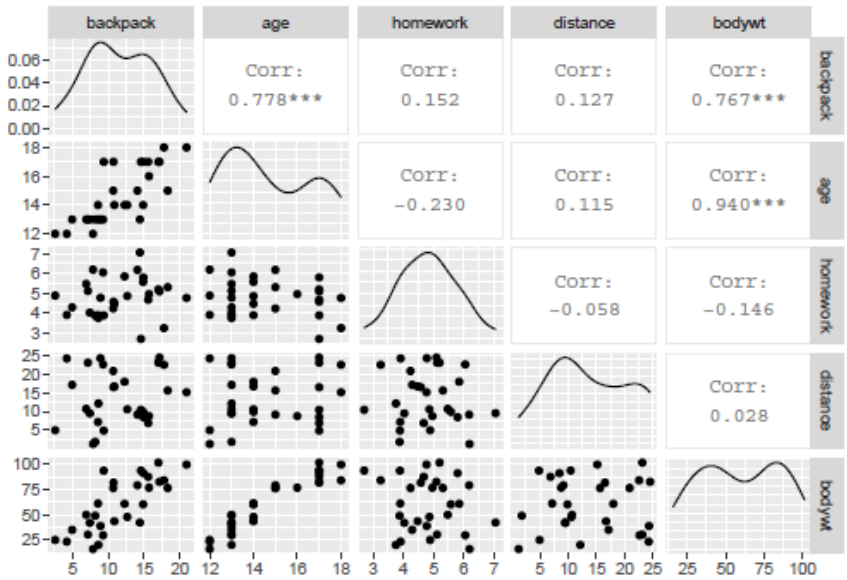
Question 3

2 out of 2 points

Researchers running the School Years Welfare Study were interested in explaining the variability in weights (kgs) of backpacks carried by children, variable backpack. The following candidate predictor variables were recorded for a sample of 30 children:

- age : the age of the child in years
- bodywt : bodyweight in kgs.
- homework : number of hours homework per week
- distance : distance between home and school in kilometres.

Descriptive statistics and scatterplots for the quantitative variables are provided below.



The sample Pearson's correlation coefficient between backpack and age is reported to be r = 0.778, **interpret** the meaning of this value.

Selected Answer: Since this is close to 1 in magnitude and positive in value this indicates a strong positive linear relationship between the weight of backpacks and age of children in the sample.

Response Feedback: a strong increasing linear

Question 4

6 out of 6 points

Carry out Pearson's correlation test to determine whether there is evidence of a significant increasing relationship between the variables backpack and age in the population. Include clear statements of the hypotheses, the rejection region and conclusion.

Selected Answer:

Null hypothesis H_0: $\rho = 0$ versus Alternative Hypothesis H_a: $\rho > 0$

Test Statistic = $r\sqrt{\frac{n-2}{1-r^2}} = 0.778\sqrt{\frac{30-2}{1-0.778^2}} = 6.5526$

Sampling distribution: $t_{n-2} = t_{30-2} = t_{28}$. Rejection region for the test statistic is: greater than $t_{28,0.05} = 1.701$.

Since the TS = 6.5526 is in this rejection region there is evidence to reject H_0 and suggest H_a at level $\alpha = 0.05$.
Alternatively, the p-value is calculated at $P(t_{28} > 6.5526) = 0.0000002096527$.
There is evidence in the sample to suggest an increasing linear relationship exists between the weight of backpack carried by children and the age of the children in the population of all children.

Response Feedback: [None Given]

Question 5

1 out of 1 points

Which of the following is a non-parametric alternative test to the Pearson's correlation test?

Selected Answer:

Spearman's Rank test

Question 6

1 out of 2 points

A multiple linear regression model, using the three candidate predictors is proposed:
$backpack_i = \beta_0 + \beta_1 age_i + \beta_2 homework_i + \beta_3 distance_i + \varepsilon_i$

What does the error term, ε_i, represent?

Selected Answer: The error term ε_i represents the random error unaccounted for in the multiple linear regression model. This is the sum of the differences between the regression line and the observed results.

Response Feedback: The error term ε_i represents the random error unaccounted for in the multiple linear regression model. ~~This is the sum of the differences between the regression line and the observed results.~~

the deviation in response for individual i away from the overall general population trend

Question 7

5 out of 5 points

When carrying out an Analysis of Variance for any regression analysis, **explain in words** what the terms SSTotal, SSReg and SSRes represent.

Selected Answer:

SS_{Total} represents the total variability found in the response variable, and is measured by the sums of squared difference between the observed values of the response variable and its overall mean, SSyy = $\sum(y_i - \bar{y}_i)^2$

SS_{Reg} represents the variability in the observed responses that is explained by our regression model, and is measured by the sums of squared difference between the predicted values of the response variable and its overall mean, $\sum(\hat{y}_i - \bar{y}_i)^2$

SS_{Res} represents the variability in the observed responses not explained by our regression model, due to unaccounted for random/residual error, and is measured by the sums of squared difference between the observed values of the response variable and the regression model's predicted values, $\sum(y_i - \hat{y}_i)^2$

Response Feedback: *(formulae not required)*

Question 8

4 out of 4 points

The model, $backpack_i = \beta_0 + \beta_1 age_i + \beta_2 homework_i + \beta_3 distance_i + \varepsilon_i$, is fitted to the observations in the sample of 30 individuals and the results of the fit are provided in the partial printout below.

```
Coefficients:
              Estimate  Std. Error   t value   Pr(>|t|)
(Intercept)   -26.2531    4.8922    -5.3663    0.0000
age             2.0009    0.2503    7.9929    0.0000
homework        1.6775    0.5071      -          -
distance        0.0318    0.6666    0.4775    0.6370
---
Residual standard error:  -  on  -  degrees of freedom
Multiple R-squared:   -   ,    Adjusted R-squared:  -
F-statistic: 22.6  on  -  and  -  DF,  p-value: 0.0000002054334
```

Analysis of variance for the model fitted to the observed data gives the values SS_{Total} = 611.18 and SS_{Reg} = 441.78.

i) The coefficient of determination R^2 is missing from the output. Find this value.

[answerii]

ii) The value of the residual standard error, s, is missing from the output. Find this value.

[answeriii]

Specified Answer for: answerii	0.72
Specified Answer for: answeriii	2.55

Question 9

2 out of 2 points

Interpret the meaning of the value of the coefficient of determination, R^2, in this application.

Selected Answer: The coefficient of determination for this model with three predictor variables is 0.72, i.e. 72% of the variability observed in the response variable backpack weight in the sample, is explained by the model with predictors age, homework and distance.

Response [None Given]

Feedback:

Question 10

3 out of 3 points

Give an interpretation for the fitted coefficient for each of the three predictor variables.

Selected Answer: Coefficient for age: for each year increase of a child's age (age), the weight of their backpack increases by 2.0009 kg on average, holding the other predictor variables (homework, distance) constant.

Coefficient for homework: for each hour increase in hours of homework per week a child has(homework), the weight of their backpack increases by 1.6775 kg on average, holding the other predictor variables (age, distance) constant.

Coefficient for distance: for each kilometre increase in distance between a child's home and their school, the weight of their backpack increases by 0.0318 kg on average, holding the other predictor variables (age, homework) constant.

Response Feedback: [None Given]

Question 11

6 out of 6 points

The test statistic for the ANOVA F-test is printed as 22.6. Show how this value is calculated.

Write down the hypotheses being tested.

Referencing the p-value provided, what is the conlcusion for this test in terms of the variables in the application.

Selected Answer: The ANOVA F-test test statistic is the ratio $\frac{MS_{Reg}}{MS_{Res}}$.

Degrees of Freedom:

df_{Total} = n - 1 = 30 - 1 = 29

df_{Reg} = number of predictors in the model = 3

$df_{Res} = df_{Total} - df_{Reg}$ = 29 - 3 = 26

Mean Squares:

$MS_{Reg} = \frac{SS_{Reg}}{df_{Reg}} = \frac{441.78}{3} = 147.26$

$MS_{Res} = \frac{SS_{Res}}{df_{Res}} = \frac{169.4}{26} = 6.52$

Test Statistic $= \frac{MS_{Reg}}{MS_{Res}} = \frac{147.26}{6.52} = 22.59 = 22.6$

Null Hypothesis H_0: the model, with predictors age, homework and distance, does not explain the variability in the response backpack, $\beta_1 = \beta_2 = \beta_3 = 0$

Alternative Hypothesis H_a: the model, with predictors age, homework and distance, does explain the variability in the response backpack, at least one of $\beta_1 \neq 0$, $\beta_2 \neq 0$, $\beta_3 \neq 0$.

The p-value of 2.054334×10^{-7} is small, there is evidence to reject the null hypothesis, there is evidence to suggest Ha: the model with age, homework, and distance is explaining the variability in children's backpack weight in the population, i.e. at least one good predictor in the model, at least one of $\beta_1 \neq 0$, $\beta_2 \neq 0$, $\beta_3 \neq 0$.

Response Feedback: [None Given]

Question 12

5 out of 5 points

Complete the missing t-test in the output, i.e. test whether homework is a good predictor of backpack at the α = 0.05 level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion.

Selected Answer: Null Hypothesis H_0: $\beta_2 = 0$ versus Alternative Hypothesis H_a: $\beta_2 \neq 0$

Sample statistics: n = 30, $\widehat{\beta_2}$ = 1.6775

Standard error: 0.5071

Test Statistic is: $\frac{\widehat{\beta_2} - \beta_2}{SE} = \frac{1.6775 - 0}{0.5071}$ = 3.308

Distribution of the test statistic: $t_{dfRes} = t_{26}$

Two tailed test: P-value = $2P(t_{26} > 3.308) = 2(0.001293495) = 0.002586991$

$$\widehat{\beta_2} \pm t_{26,0.025}SE$$

1.6775 ± 2.056(0.5071) = 1.6775 ± 1.0425976

$0.63 < \beta_2 < 2.72$

The p-value is small, there is evidence to reject H_0, evidence to suggest H_a. There is evidence to suggest, $\beta_2 \neq 0$, the effect of hours of homework per week on backpack weight is significant in the population.

The effect of a 1 hour increase in hours of homework per week is an increase in backpack weight on average, estimated to be between 0.63 and 2.72 kg

Response Feedback: *confidence interval for B2 not requested here.*

interpretation of the effect was not requested here.

Question 13

3 out of 3 points

What can you conclude as a result of the t-tests in this output?
Choose one or more from the following:

Selected Answers:

The t-test for $H_0 : \beta_2 = 0$ versus H_a: $\beta_2 \neq 0$ suggests there is evidence that homework is a good predictor of backpack weight.

The t-test for $H_0 : \beta_1 = 0$ versus H_a: $\beta_1 \neq 0$ suggests there is evidence that age is a good predictor of backpack weight.

The t-test for $H_0 : \beta_3 = 0$ versus H_a: $\beta_3 \neq 0$ suggests there is no evidence that distance is a good predictor of backpack weight.

Question 14

0 out of 2 points

The output below provides a prediction interval and a confidence interval for the response when age = 14, homework = 5 and distance = 11.

Prediction interval for the response:

```
       fit      lwr      upr
1 10.49695 5.142898 15.85099
```

Confidence interval for the response:

```
       fit      lwr      upr
1 10.49695 9.430513 11.56338
```

Explain, referencing the application, the difference as to when a prediction interval and a confidence interval applies.

Selected Answer:	A confidence interval applies when you want to estimate estimate the range of values where the line of best fit or mean of the data will lie. A prediction interval is a wider range, in which 95% of the response variable data points will lie.
Response Feedback:	Explain, referencing the application, see solutions.

Question 15

2 out of 2 points

In one sentence, explain the term "multicollinearity".

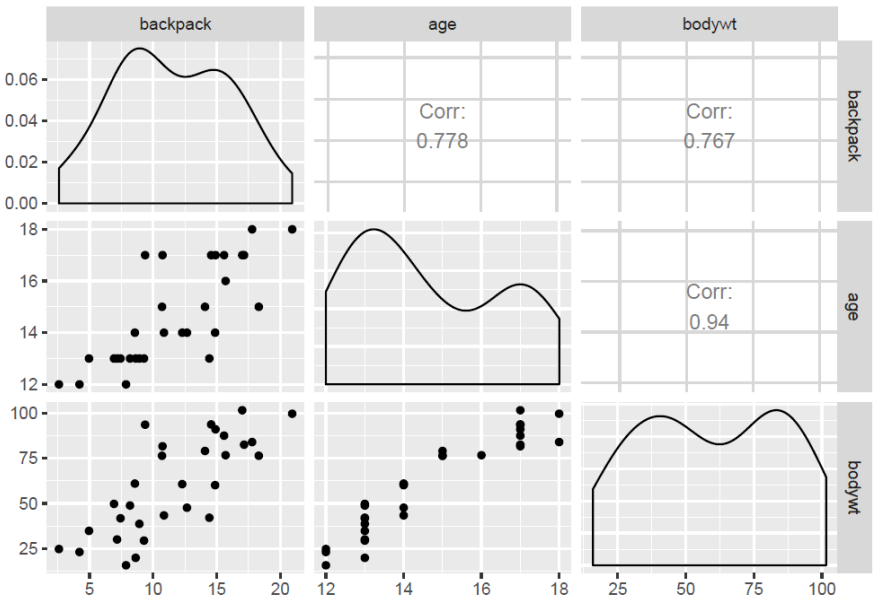
Selected Answer:	We say that a set of predictors exhibits multicolinearity when one or more of the predictors is strongly correlated with some combination of the other predictors in the set.
Response Feedback:	[None Given]

Question 16

9 out of 10 points

The plot and output below is produced to assess a model where age and bodywt are used to predict backpack weight,

$\text{backpack}_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{bodywt}_i + \varepsilon_i$.



```
Call:
lm(formula = backpack ~ age + bodywt, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.6914 -1.9923  0.4405  1.6110  5.6846 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.42285    8.70075  -0.968    0.342
age          1.14748    0.82034   1.399    0.173
bodywt       0.05324    0.06059   0.879    0.387

Residual standard error: 2.949 on 27 degrees of freedom
Multiple R-squared:  0.6157,	Adjusted R-squared:  0.5873 
F-statistic: 21.63 on 2 and 27 DF,  p-value: 0.00000247
```

Discuss the predictive power of the two variables, age and bodywt, by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference.

Selected Answer: The t-tests in the output for each predictor is used to determine whether the coefficient of the predictor variables age and bodywt are significantly different from 0, i.e. H_0: $\beta_1 = 0$ versus H_a: $\beta_1 \neq 0$

The t-test for the coefficient of age gives a test statistic of 1.399 and p-value 0.173 The p-value is large, there is no evidence to reject the null hypothesis, no evidence to suggest that $\beta_1 \neq 0$ there is no evidence to suggest that age is a good predictor of the backpack weight in the population

The t-test for the coefficient of bodywt gives a test statistic of 0.879 and p-value 0.387 The p-value is large, there is no evidence to reject the null hypothesis, no evidence to suggest that $\beta_2 \neq 0$ there is no evidence to suggest that bodywt is a good predictor of the average backpack weight in the population.

The Anova test:
Null Hypothesis H_0: the model, with predictors age and bodywt, does not explain the variability in the response backpack, $\beta_1 = \beta_2 = 0$

Alternative Hypothesis H_a: the model, with predictors age and bodywt, does explain the variability in the response backpack, at least one of $\beta_1 \neq 0$, $\beta_2 \neq 0$, $\beta_3 \neq 0$.

The ANOVA F-test statistic has value 21.63 with a p-value of 0.00000247. The p-value is small, there is evidence to reject the null hypothesis, there is evidence to suggest the model with age and bodywt is explaining the variability in backpack weight in the population, i.e. coefficient of at least one of the predictor variables, age and bodywt, has a coefficient $\neq 0$ in the population.

This contradiction between the ANOVA and t-tests, where our ANOVA test suggests that at least one of age and bodywt is a good predictor of backpack weight while each of their t-tests suggest no evidence that either are good predictors, is due to to multicollinearity. The correlation present between the predictors interferes with the outcome of our t-tests.

Response Feedback: This contradiction between the ANOVA and t-tests, where our ANOVA test suggests that at least one of age and bodywt is a good predictor of backpack weight while each of their t-tests suggest no evidence that either are good predictors, is due to to multicollinearity. The correlation present between the predictors interferes with the outcome of our t-tests....

...correct, and you can add...

the observed correaltion between age and bodywt is 0.94 indicating a strong increasing linear relationship between the predictors.

Question 17

2 out of 2 points

In one sentence, explain the meaning of the phrase "a parsimonious model" in variable selection.

Selected Answer: A parsimonious model is the simplest best fitting model, the best fitting model in which the number of explanatory variables is kept to a minimum.

Response Feedback: [None Given]

Question 18

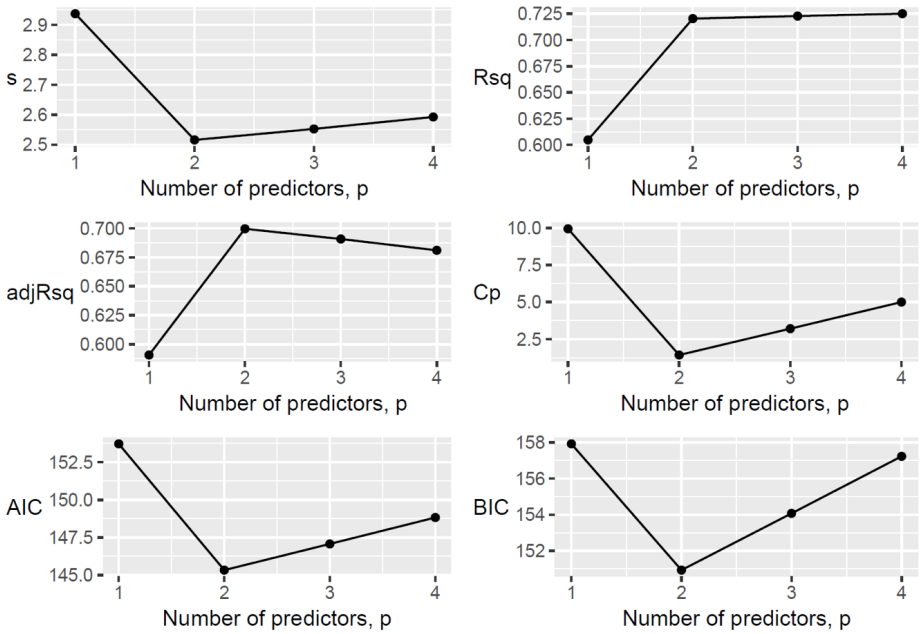
8 out of 8 points

A best-subsets routine was carried out on 4 candidate predictors, age, homework, distance and bodywt.

The best fitting model for each number of predictors is selected and summarised below.

model	predictors	p	s	Rsq	adjRsq	Cp	AIC	BIC
1	age	1	2.937	0.605	0.591	9.944	153.716	157.920
2	age homework	2	2.516	0.720	0.700	1.426	145.330	150.935
3	age homework distance	3	2.553	0.723	0.691	3.204	147.068	154.074
4	age homework distance bodywt	4	2.592	0.725	0.681	5.000	148.824	157.231

.



Using the output to support your answer, write down the most suitable model, i.e. the set of predictors, that you think best explains the variability in observed backpack. Justify your choice of model by explaining the role of s, R^2, adjusted R^2, and Mallow's Cp in choosing the best model.

Selected Answer:

The most suitable model, the model that best fits the observed data, is achieved when the residuals are minimised, and therefore when the Residual Sum of Squares is minimised. The residual standard error can be used to investigate this, as the residual standard error s = $\sqrt{\frac{\sum \hat{e}_i^2}{n-p-1}} = \sqrt{\frac{SS_{Res}}{df_{Res}}} = \sqrt{MS_{Res}}$. We can see that minimising s will also minimise the residual sum of squares, while also incorporating the number of predictors in the calculation (df_{Res}).
s is minimised for model 2 at a value of 2.516.

R-squared, the coefficient of determination, is the proportion of variability explained in the response by the predictors in each model. We want to maximise this to find the most suitable model, however R-squared will always increase as we add variables. Adjusted R-squared adds a penalty term dependent on the number of predictors in the model, and maximising this instead of R-squared will ensure that the the best fitting model is chosen.

Adjusted R-squared is maximised at 0.700 in model 2, with predictors age and homework.

Mallow's C_p should be less than or equal to the number of parameters in the model equation, and this is true for model 2 only: $1.426 \leq 2$

Akaike's Information Criterion (AIC) and Bayesian Information Criterion (BIC) should be minimised and both are minimised for model 2, at 145.330 and 150.935 respectively.

It is evident from these observations that model 2, with predictors age and homework, is the most suitable model.

Response Feedback: [None Given]

Question 19

3 out of 4 points

When fitting a linear regression model to data what assumptions are required?

Selected Answer: Linear regression requires us to make assumptions about:

Linearity: the relationship between predictor variable X and the mean of response variable Y is linear.
Homoscedascticity: the variance of the residuals are constant.
Independence: the residuals are independent
Normality: the residuals are normally distributed.

Response Feedback: Linear regression requires us to make assumptions about:

~~Linearity: the relationship between predictor variable X and the mean of response variable Y is linear.~~

Homoscedascticity: the variance of the residuals are constant.

Independence: the residuals are independent

Normality: the residuals are normally distributed.

Residuals have expectation 0

Question 20

3 out of 3 points

The percentage load of weight of the backpack, PercLoad, is calculated as the backpack weight relative to the individual's body mass.

The following output outlines the fit for the predictor age on the response PercLoad,

$PercLoad_i = \beta_0 + \beta_1 age_i + \varepsilon_i$

```
Call:
lm(formula = PercLoad ~ age, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.7928 -1.3114 -0.6993  1.1469  9.1218 

Coefficients:
            Estimate Std. Error t value   Pr(>|t|)    
(Intercept)  23.2933     4.2454   5.487 0.00000735 ***
age          -0.9826     0.2876  -3.416    0.00196 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.03 on 28 degrees of freedom
Multiple R-squared:  0.2942,	Adjusted R-squared:  0.269 
F-statistic: 11.67 on 1 and 28 DF,  p-value: 0.001959
```

i) Use the estimated regression equation for this model to find the predicted value of PercLoad for a child aged 12 years old.

[pred]

ii) The observed PercLoad for this child was 20.62. Calculate the fitted residual.

[res]

iii) Find the value of the standardised residual for this observation.

[stres]

Specified Answer for: pred	11.5
Specified Answer for: res	9.12
Specified Answer for: stres	3

Question 21

2 out of 2 points

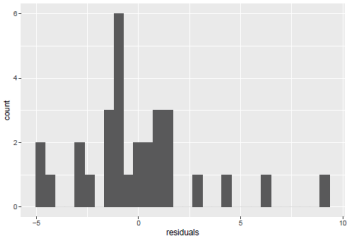
Explain what assumption the standardised residuals are used to validate. What would you say about the observation in the previous question?

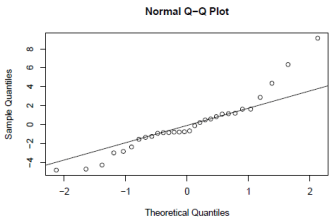
Selected Answer:	Standardised residuals can be used to validate normality assumptions, and help identify any outliers. 95% of the standardised residuals should fall between -2 and 2. I would deem the standardised residual value of 3 in our observation to indicate that this data point is an outlier.
Response Feedback:	[None Given]

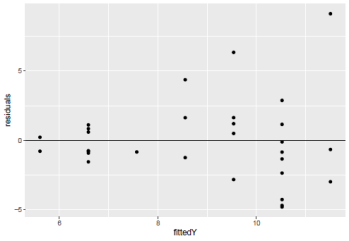
Question 22

4 out of 4 points

Diagnositc Plots for the fitted regression model for PercLoad are produced:







Explain how the diagnostic plots are used to assess validity of the assumptions. Comment on the outputs of these diagnostic plots.

Selected Answer: Both the histogram and the Normal Q-Q Plot can be used to assess the normality of residuals assumption. The Residuals vs Fitted plot can be used to asses constant variance of residuals homoscedasticity assumption.

The Normal Q-Q Plot shows the observed rougly lying along the diagonal line of equality, while there is deviation in the tails there is not strong evidence of departure. This suggests a normal distribution.

The Residual vs Fitted plot shows that residuals have an expectation value of 0. The pattern exhibited by the plot, however, indicates heteroscedasticity in the data, the variance of the residuals is increasing as our observation value increases, rather than the randomly scattered constant variance required in our assumption of homoscedasticity.

Response Feedback: [None Given]

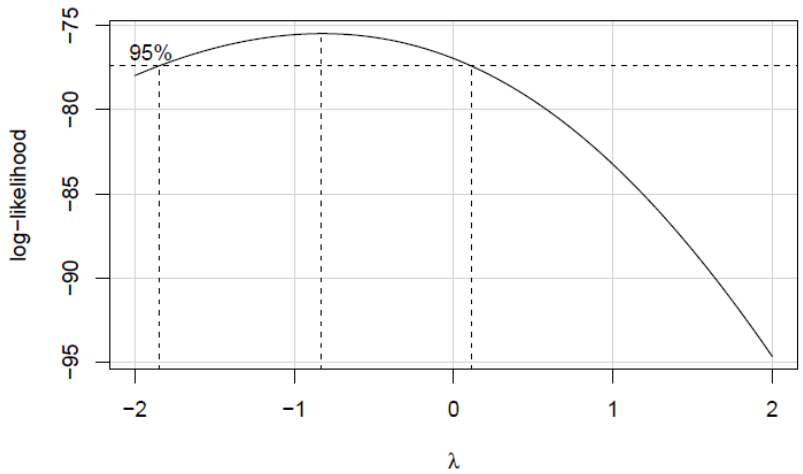
Question 23

4 out of 4 points

What does the following box-cox transformation output indicate?

```
bcPower Transformation to Normality
   Est Power Rounded Pwr Wald Lwr bnd Wald Upr Bnd
Y1   -0.8236           0       -1.798       0.1507

Likelihood ratio tests about transformation parameters
                           LRT df         pval
LR test, lambda = (0)  2.959074  1 0.0853968227
LR test, lambda = (1) 15.545215  1 0.0000805553
```



Selected Answer:

The Likelihood ratio test for H_0: $\lambda = 1$ versus H_a: $\lambda \neq 1$ has p-value 0.0000805553, which is small giving evidence to reject H_0 , evidence to suggest H_a: $\lambda \neq 1$, therefore suggests a transformation is required.

The Likelihood ratio test for H_0: $\lambda = 0$ versus H_a: $\lambda \neq 0$ has p-value 0.0853968227, which is not small giving evidence to reject H_0 , therefore a log transformation may be applicable.

Alternatively:
The confidence interval for the value of λ, the power on the response variable, Y^{λ}, is: $-1.798 < \lambda < 0.1507$. This interval doesn't contain $\lambda = 1$ therefore suggests a transformation is required. This interval does contain $\lambda = 0$ therefore suggests a log transformation.

Response Feedback: [None Given]

Question 24

4 out of 4 points

A simple linear regression using age as the predictor was fitted to the transformed response variable ln(PercLoad) ,

$\ln(\text{PercLoad})_i = \beta_0 + \beta_1 \text{age}_i + \varepsilon_i$

giving the output below.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.7062	0.4123	8.9886	<0.0001
age	-0.1083	0.0279	-3.8752	0.0006

Using this model **predict** the percentage load of weight of the backpack of an individual aged 12 years.

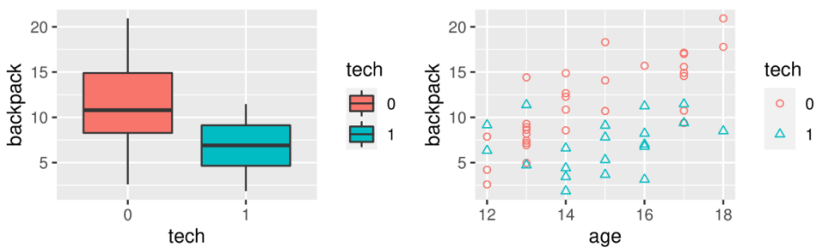
Selected Answer: 11.1

Question 25

2 out of 2 points

Researchers wished to assess the effect on backpack weight when children use technology in their school day, e.g. iPad/tablet. Backpack weight was observed for children who do use technology and for children who don't use technology in the school day, the variable tech is stored as an indicator variable defined as

- $\text{tech}_i=0$, if child i uses no technology and
- $\text{tech}_i=1$, if child i uses technology.



A model using the indicator variable tech, i.e. $\text{backpack}_i = \beta_0 + \beta_1 \text{tech}_i + \varepsilon_i$, is fitted to the data giving the output below.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	11.5554	0.7303	15.8233	<0.0001
tech	-4.5821	1.1547	-3.9683	0.0002

Using this fitted model provide the estimate of backpack weight for:

- a child that doesn't use technology.
- a child that does use technology.

Selected Answer: $\widehat{\text{backpack}}_i = 11.5554 - 4.5821\text{tech}_i$

A child doesn't use technology, tech = 0:
backpack weight is estimated to be 11.5554 units.

A child that does use technology, tech = 1:
backpack weight is estimated to be 11.5554 - 4.5821 units = 6.9733 units.

Response Feedback: [None Given]

Question 26

3 out of 3 points

Interpret the meaning of the estimated coefficient of tech and comment on the significance of the effect. Ensure you state the hypotheses being tested when referring to the results of the output.

Selected Answer: Coefficient of the indicator variable tech, $\widehat{\beta_1} = -4.5821$: this is the difference on average in backpack weight between children who use technology and children who don't use technology - i.e. the backpack weight of children who use technology is 4.5821 units less on average than the backpack weight of children who don't use technology.

The t-test in the ouput is used to determine whether the coefficient of the indicator variable tech is significantly different from 0, i.e. H_0: $\beta_1 = 0$ versus H_a: $\beta_1 \neq 0$

The p-value for this statistic is small, with $p < 0.0002$, giving evidence to reject H_0. There is evidence to suggest H_a: $\beta_1 \neq 0$. There is evidence to suggest that true mean backpack weight for the population of technology using children and non-technology using children are different.

Response Feedback: [None Given]

Question 27

1 out of 1 points

This analysis is equivalent to which of the following two population comparison tests:

Selected Answers: Independent sample t-test for difference in population means with equal variance

Question 28

3 out of 3 points

A plot of the observed relationship of backpack weight and the age of the child was also provided, with an indication to those who did and didn't use tech.

Age is added to the model as a covariate of predictor tech, in the model

$\text{backpack}_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{tech}_i + \varepsilon_i$

and fitted to the data giving the following output.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-8.6823	3.6202	-2.3983	0.0205
age	1.3830	0.2443	5.6606	<0.0001
tech	-4.9509	0.9022	-5.4878	<0.0001

Using this fitted model provide the estimated relationship between backpack and age for:

- a child that doesn't use technology.

- a child that does use technology.

Selected Answer:

$$\widehat{\text{backpack}}_i = -8.6823 + 1.383\text{age}_i - 4.9509\text{tech}_i$$

A child doesn't use technology, $\text{tech}_i = 0$:
Estimated backpack weight = -8.6823 + 1.383age_i - 4.9509(0) = -8.6823 + 1.383age_i
backpack = -8.6823 + 1.383age_i
A child who does use technology, $\text{tech}_i = 1$:
Estimated backpack weight = -8.6823 + 1.383age_i - 4.9509(1) = -13.6332 + 1.383age_i
backpack = -13.6332 + 1.383age_i

Response Feedback: [None Given]

Question 29

4 out of 4 points

Interpret the meaning of the estimated coefficients of age and tech and comment on the significance of the effects. Ensure you state the hypotheses being tested when referring to the results of the output.

Selected Answer: Coefficient of the predictor variable age: β_1 = 1.3830: for each additional year in a child's age, the weight of their backpack increases by 1.3830 units on average.

Coefficient of the predictor variable tech: β_2 = -4.9509: this is the difference on average in backpack weight between children who don't use technology and children who do, i.e. the weight of backpacks of children who use technology are less on average than children who don't use technology bby 4.9509 units. This difference is assumed the same for all ages of children.

The t-tests in the ouput are used to determine whether the coefficient of the predictor variables age and tech are significantly different from 0, i.e. H_0: $\beta_1 = 0$ versus H_a: $\beta_1 \neq 0$

The p-value of the t-test for the predictor variable age statistic is small, with $p < 0.0001$, giving evidence to reject H_0. There is evidence to suggest H_a: $\beta_1 \neq 0$ i.e. the effect of children's age on the weight of their backpack is significant in the population.

The p-value of the t-test for the predictor variable tech is small, with $p < 0.0001$, giving evidence to reject H_0. There is evidence to suggest H_a: $\beta_1 \neq 0$, evidence of a difference in children's backpack weight comparing those who use technology and those who don't in the population.

Response Feedback: [None Given]

Question 30

2 out of 2 points

What assumption is being made in this model about the relationship between backpack weight and age when tech is used or not used?

Selected Answer: This model assumes that the change in backpack between children who don't use and do use technology remains constant regardless of the age of the children. This model assume equal slopes comparing the populations.

Response Feedback: [None Given]

Question 31

4 out of 4 points

An interaction term of age and tech, defined as age:tech$_i$=(age$_i$) x (tech$_i$), is added to the model

backpack$_i$ = $\beta_0 + \beta_1$age$_i$ + $\beta_2$tech$_i$ + β_3 age:tech$_i$ +ε_i

and fitted to the data giving the following output.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-15.1509	4.0718	-3.7210	0.0005
age	1.8250	0.2759	6.6153	<0.0001
tech	15.6276	7.2938	2.1426	0.0375
age:tech	-1.3890	0.4890	-2.8403	0.0067

Using this fitted model provide the estimated relationship between backpack weight and age for:

- a child that doesn't use technology.
- a child that does use technology.

Selected Answer:

$\widehat{\text{backpack}}_i = -15.1509 + 1.825\text{age}_i + 15.6276\text{tech}_i - 1.389(\text{age}_i \text{x tech}_i)$

A child doesn't use technology, tech$_i$ = 0:
Estimated backpack weight = -15.1509 + 1.825age$_i$ + 15.6276(0) - 1.389(age$_i$ x 0) = -15.1509 +1.825age$_i$
backpack = -15.1509 +1.825age$_i$
A child who does use technology, tech$_i$ = 1:
Estimated backpack weight = -15.1509 + 1.825age$_i$ + 15.6276(1) - 1.389(age$_i$ x 1) = 0.4767 + 0.436age$_i$
backpack = 0.4767 + 0.436age$_i$

Response Feedback: [None Given]

Question 32

5 out of 5 points

Interpret the meaning of the fitted coefficient of age : tech in this model. Determine if the term is significant, using the result of the individual t-test in this analysis. Ensure you state the hypotheses being tested when referring to the results of the output.

Selected Answer: Coefficient of the interaction term age:tech, -1.3890: difference in slopes comparing backpack weight between children who use technology and children who don't, 1.8250 - 1.3890 = 0.436 : for children who use technology, for each additional year of age backpack weight increases by 0.436 units on average. This increase in backpack weight per year of age is 1.3890 less than the increase in backpack weight per year of age for children who don't use technology.

Testing for non zero coefficient of the coefficient of the interaction term age:tech, the estimated effect is -1.3890, the test statistic is -2.8403, with p-value 0.0067. There is evidence to reject null hypothesis, there is evidence to suggest the coefficient of the interaction term is not equal to 0, evidence of a difference in slopes in the relationship of children's backpack weight and children's age comparing those that use technology and those that do not in the population.

Response Feedback: [None Given]

Monday, 4 January 2021 13:07:28 o'clock GMT

OK